

Processando config 9, seed 0, teste: Rabbit
XX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

smultin_MultinomialBitsOver test:

N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE

Number of bits = n = 8388608
Number of cells = 2^L = 274877906944
Expected number per cell = $1 / 32768$
EColl = $n^2 / (2k)$ = 128
Hashing = TRUE

Collision test
CollisionOver: density = $n / k = 1 / 32768.00$
Expected number of collisions = $\mu = 128.00$

Results of CollisionOver test:

POISSON approximation :
Expected number of collisions = $N \cdot \mu$: 256.00
Observed number of collisions : 219
p-value of test : 0.9900

Total number of cells containing j balls

j = 0	:	549739036891
j = 1	:	16776778
j = 2	:	219
j = 3	:	0
j = 4	:	0
j = 5	:	0

CPU time used : 00:00:04.44

Generator state:
16777216 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 16.00
p-value of test : 0.86

CPU time used : 00:00:00.06

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 9.00
p-value of test : 7.8e-3

CPU time used : 00:00:00.06

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : -0.69
p-value of test : 0.76

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 10.43
p-value of test : 0.17

Normal statistic for number of jumps : 1.25
p-value of test : 0.11

CPU time used : 00:00:00.18

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : 0.24
p-value of test : 0.41

CPU time used : 00:00:00.40

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : 0.50
p-value of test : 0.31

CPU time used : 00:00:00.21

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.036
p-value of test : 0.26

Kolmogorov-Smirnov- statistic = D- : 0.020
p-value of test : 0.67

Anderson-Darling statistic = A2 : 0.53
p-value of test : 0.72

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 3.99
p-value of test : 0.26

Global longest run of 1 : 23.00
p-value of test : 0.50

CPU time used : 00:00:00.06

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 92.03
p-value of test : 0.54

CPU time used : 00:00:00.00

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 23.68
p-value of test : 0.36

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -1.26
p-value of test : 0.90

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : -0.25
p-value of test : 0.60

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : -0.30
p-value of test : 0.62

CPU time used : 00:00:00.02

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 154.00
p-value of test : 0.89

CPU time used : 00:00:00.00

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 325.71
p-value of test : 0.20

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 485.61
p-value of test : 0.53

CPU time used : 00:00:00.02

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          :   -0.28
p-value of test          :    0.61
-----
```

```
CPU time used            :   00:00:00.00
```

```
Generator state:
 269499904 bits have been read.
```

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*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin
```

```
sstring_AutoCor test:
```

```
-----
      N =  1,   n = 16777154,   r =  0,   s = 32,   d = 2
-----
```

```
-----
Normal statistic          :    1.69
p-value of test          :    0.05
-----
```

```
CPU time used            :   00:00:00.02
```

```
Generator state:
 286277088 bits have been read.
```

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*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin
```

```
sstring_Run test:
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```
-----
      N =  1,   n = 3355443,   r =  0,   s =  32
-----
```

```
-----
Total number of 1 runs:  3355451
-----
```

```
Number of degrees of freedom      :   36
Chi2 statistic for number of runs :  30.01
p-value of test                   :    0.75
-----
```

```
-----
Total number of bits:  13418912
-----
```

```
Normal statistic for number of bits :   -0.55
p-value of test                   :    0.71
```

CPU time used : 00:00:00.03

Generator state:
299696000 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 3.60
p-value of test : 0.31

CPU time used : 00:00:00.06

Generator state:
316473216 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.09

Generator state:
333164416 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42
ChiSquare statistic	:	41.15
p-value of test	:	0.51

Test on the values of the Statistic M

Number of degrees of freedom	:	42
ChiSquare statistic	:	53.96
p-value of test	:	0.10

Test on the values of the Statistic J

Number of degrees of freedom	:	64
ChiSquare statistic	:	58.01
p-value of test	:	0.69

Test on the values of the Statistic R

Number of degrees of freedom	:	36
ChiSquare statistic	:	46.47
p-value of test	:	0.11

Test on the values of the Statistic C

Number of degrees of freedom	:	22
ChiSquare statistic	:	19.82
p-value of test	:	0.59

CPU time used : 00:00:00.05

Generator state:
349941632 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94
ChiSquare statistic	:	79.13
p-value of test	:	0.86

Test on the values of the Statistic M

Number of degrees of freedom	:	94
ChiSquare statistic	:	97.64
p-value of test	:	0.38

Test on the values of the Statistic J

Number of degrees of freedom	:	512
ChiSquare statistic	:	598.75
p-value of test	:	4.8e-3

Test on the values of the Statistic R

Number of degrees of freedom	:	90
ChiSquare statistic	:	89.98
p-value of test	:	0.48

Test on the values of the Statistic C

Number of degrees of freedom	:	50
ChiSquare statistic	:	63.75
p-value of test	:	0.09

CPU time used : 00:00:00.05

Generator state:

366718848 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config9_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	120.85
p-value of test	:	0.71

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	146.71
p-value of test	:	0.16

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	162.03
p-value of test	:	0.46

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	120.73
p-value of test	:	0.71

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	83.95
p-value of test	:	0.63

CPU time used : 00:00:00.03

Generator state:

383495648 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.12
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
22 MatrixRank, 320 x 320	eps
-----	-----

All other tests were passed